Huawei Access Node KR-ACC-72 Optical Transport Failure Resolution SOP (Severity: Critical)

Node: KR-ACC-72 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5425

Description

The Optical Transport Failure (OTF) failure domain in Huawei Access Nodes is a critical failure condition characterized by the degradation of SFP/XFP interfaces, fiber cuts, or Rx/Tx power loss. This failure domain occurs due to the underlying protocol and hardware mechanics of the Huawei Access Node, specifically the OLT (Optical Line Terminal) and ONU (Optical Network Unit) interfaces. In the Huawei Access Node, the OLT and ONU interfaces use the GPON (Gigabit Passive Optical Network) protocol to transmit and receive data over the optical fiber network. The GPON protocol relies on the integrity of the optical signal to transmit data, and any degradation or loss of the optical signal can cause a failure in the OTF. In the event of a fiber cut or Rx/Tx power loss, the OLT and ONU interfaces will detect the failure and raise an alarm, triggering the OTF failure domain. To resolve this failure, it is essential to identify and address the root cause of the failure, whether it be a hardware issue, a software issue, or a physical infrastructure issue. This SOP outlines the diagnostic commands, resolution steps, and verification commands required to resolve the OTF failure domain in the Huawei Access Node KR-ACC-72.

Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and that all necessary personnel and equipment are present.

- Check the redundant link status to ensure that it is operational and can take over in the event of a failure.

- Confirm that the Huawei Access Node KR-ACC-72 is properly powered on and that all necessary alarms are silenced.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display device sfp, Expected Output: SFP information: SFP-1: present, SFP-2: present, Description: This command displays the status of the SFP/XFP interfaces on the Huawei Access Node. The expected output indicates that both SFP-1 and SFP-2 are present, suggesting that the SFP/XFP degradation is not the root cause of the OTF failure
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domain.

Command: display optical-transport interface, Expected Output:
Optical-transport interface: Optical-1: operational, Description: This
command displays the status of the optical transport interface on the
Huawei Access Node. The expected output indicates that the Optical-1
interface is operational, suggesting that the OTF failure domain is
not due to a fiber cut.

Command: display rx-tx-power, Expected Output: Rx/Tx power: Rx: -20
dBm, Tx: -5 dBm, Description: This command displays the Rx/Tx power
levels on the Huawei Access Node. The expected output indicates that
the Rx/Tx power levels are within the normal range, suggesting that
the OTF failure domain is not due to Rx/Tx power loss.
```

Resolution Steps

Step 1: Step: Replace faulty SFP/XFP interface, Command: replace sfp SFP-1, Description: This command replaces the faulty SFP/XFP interface on the Huawei Access Node. The replacement SFP/XFP interface should be properly configured and tested to ensure that it is operational.

Step 2: Step: Reconnect and test the fiber link, Command: connect optical-transport interface Optical-1, Description: This command reconnects the optical transport interface on the Huawei Access Node. The fiber link should be properly tested to ensure that it is operational and that the OTF failure domain has been resolved.

Step 3: Step: Adjust Rx/Tx power levels, Command: set rx-tx-power Rx -10 dBm, Description: This command adjusts the Rx/Tx power levels on the Huawei Access Node. The adjusted Rx/Tx power levels should be within the normal range to ensure that the OTF failure domain has been resolved.

Post-Check Verification

```
Command: display optical-transport interface, Expected Output:
Optical-transport interface: Optical-1: operational, Description: This
command verifies that the optical transport interface on the Huawei
Access Node is operational and that the OTF failure domain has been
resolved.

Command: display rx-tx-power, Expected Output: Rx/Tx power: Rx: -10
dBm, Tx: -5 dBm, Description: This command verifies that the Rx/Tx
power levels on the Huawei Access Node are within the normal range and
that the OTF failure domain has been resolved.
```

Roll-back Procedures

- Step: Restore original SFP/XFP interface, Command: restore sfp SFP-1, Description: This command restores the original SFP/XFP interface on the Huawei Access Node in the event that the replacement SFP/XFP interface fails to resolve the OTF failure domain.

- Step: Revert fiber link configuration, Command: revert optical-transport interface Optical-1, Description: This command reverts the fiber link configuration on the Huawei Access Node in the event that the fiber link fails to resolve the OTF failure domain.

- Step: Revert Rx/Tx power levels, Command: revert rx-tx-power Rx, Description: This command reverts the Rx/Tx power levels on the Huawei Access Node in the event that the adjusted Rx/Tx power levels fail to resolve the OTF failure domain.